Audio course on steroids during pregnancy. Episode eight. Adrenal insufficiency, stress dosing, and endocrine replacement.

Companion reading handout for simultaneous listening.
This episode changes the steroid question completely.

In fetal lung maturation, steroids are developmental therapy. In asthma, lupus, IBD, dermatology, or ITP, steroids are anti-inflammatory or immune therapy. In adrenal insufficiency, steroids are replacement.

That distinction matters because replacement is not optional in the same way. A patient who cannot mount an adequate cortisol response can deteriorate during labor, hemorrhage, infection, surgery, or severe illness. The danger is adrenal crisis.

So the clinical tension is this: do not mistake physiologic replacement for elective immunosuppression.

Pregnancy itself changes cortisol physiology. Cortisol levels rise progressively across gestation because corticosteroid-binding globulin increases and cortisol production increases. But baseline glucocorticoid replacement in adrenal insufficiency usually does not require routine pregnancy escalation. Dosing is individualized, but a common hydrocortisone pattern is ten milligrams in the morning and five milligrams in the afternoon.

Hydrocortisone makes sense because the placenta metabolizes it through eleven beta H S D two, protecting the fetus from excessive maternal replacement exposure.

Now recognition.

Adrenal insufficiency can be difficult to recognize in pregnancy because many symptoms overlap with normal gestation: fatigue, nausea, vomiting, lightheadedness, weakness, and pigmentation changes. But weight loss, hypoglycemia, salt craving, excessive hyponatremia, hyperkalemia, or concerning clinical deterioration should sharpen attention.

In an unstable patient, do not wait passively for perfect endocrine testing. Gabbe's endocrine chapter gives a practical emergency approach: if acute adrenal insufficiency is suspected, draw diagnostic blood if possible, then give hydrocortisone fifty to seventy five milligrams intravenously. Thereafter, fifty to seventy five milligrams every six to eight hours can be used during severe stress and labor.

Other obstetric chapters use related stress-dose regimens, such as hydrocortisone seventy five to one hundred milligrams intravenously every eight hours during labor and immediate postpartum coverage, with rapid postpartum taper.

The exact institutional regimen may vary. The principle does not: a patient with adrenal insufficiency or clinically important steroid suppression needs a stress-dose plan before stress arrives.

Now distinguish primary adrenal insufficiency from steroid-induced adrenal suppression.

A patient with Addison disease has primary adrenal insufficiency and needs replacement. A patient treated with prolonged high-dose prednisone for lupus, asthma, ITP, or another inflammatory disease may have secondary adrenal suppression. The delivery room problem can be similar: the body may not produce enough cortisol when stressed.

Gabbe's connective tissue disease chapter gives one clear threshold. Prednisone twenty milligrams per day or more for at least three weeks is a clear adrenal-suppression risk. Prednisone five milligrams per day or less has very low risk. Between five and twenty milligrams per day is a gray zone where practice varies.

Gabbe's endocrine chapter is conservative about prior anti-inflammatory glucocorticoid therapy, warning that adrenal axis suppression may persist after stopping. The bedside lesson is not to memorize one universal cutoff. The lesson is to identify risk early and plan.

Now labor.

Labor is not just a delivery event. It is physiologic stress. Cesarean delivery is stress. Hemorrhage is stress. Infection is stress. Surgery is stress. Severe vomiting and inability to take oral medication are stress.

If the patient has adrenal insufficiency, continuing her usual oral dose may be inadequate. If she is vomiting, she may not absorb oral medication at all. If she bleeds or becomes septic, cortisol need rises sharply.

A resident should ask these questions during admission, not after hypotension develops.

Does this patient have known adrenal insufficiency?

Is she on chronic glucocorticoids?

What dose and duration?

Does she need stress-dose hydrocortisone during labor or surgery?

How will the dose be tapered postpartum?

Does the neonatology team need to know about prolonged or high-dose maternal exposure?

Now compare replacement with fetal therapy.

Hydrocortisone for adrenal insufficiency is not a fetal lung maturation course. Prednisone for lupus is not a fetal lung maturation course. Betamethasone for fetal lungs is not an adrenal replacement plan.

The words are all steroids, but the clinical functions are different.

This matters postpartum too.

After delivery, steroid needs may fall quickly if the stress resolves. But abrupt withdrawal from prolonged anti-inflammatory dosing can be dangerous. Patients with true adrenal insufficiency need their baseline replacement restored. Patients with disease flares may need immunologic management. Patients who received only antenatal betamethasone for fetal maturation do not need chronic maternal taper because the regimen was brief.

Again, target determines interpretation.

Now a case.

A patient with known adrenal insufficiency arrives in labor at term. She takes hydrocortisone at home. The resident should not simply write "continue home meds" and move on. Labor is stress. There should be a stress-dose hydrocortisone plan and a postpartum taper or return-to-baseline plan.

Another patient took prednisone forty milligrams daily for several weeks for severe asthma earlier in pregnancy. She is now in labor. Even if she is no longer on that dose, review the timing and duration. She may need stress-dose planning depending on the risk of adrenal suppression and local protocol.

The resident reflex is this: do not let an adrenal-insufficient or steroid-suppressed patient labor without a stress-dose plan.

Replacement is not overmedication. It is the physiologic support the patient cannot provide.

And once again, the target-first model protects the patient. Fetal maturation, maternal inflammation, and adrenal replacement are different steroid stories. In the delivery room, confusing them can be dangerous.